

SheGuard: Resilient Offline-First Mobile Safety Architecture with Deterministic Multimodal Fusion, Cryptographic Evidence Sealing, and Peer Relay Escalation

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Abstract—Many mobile safety applications depend on manual SOS activation, continuous network connectivity, or a single detection modality, limiting their reliability when a user cannot operate a device or network access is unavailable. This paper presents SheGuard, an offline-first Android safety companion that combines on-device acoustic, keyword, and motion sensing with deterministic incident management. The system uses distress-sound classification, configurable speech-command detection, and accelerometer-based motion analysis. These signals are evaluated by a rule-based fusion engine before entering an auditable incident state machine with a bounded user-confirmation interval and default-to-escalate semantics. Following incident confirmation, SheGuard attempts emergency delivery through peer-to-peer relays over Bluetooth Low Energy or Wi-Fi Direct, with cellular-only Short Message Service fallback when relay peers or data connectivity are unavailable. In parallel, the system captures incident evidence locally, protects it using per-incident authenticated encryption, and preserves its integrity through SHA-256 hashing, Merkle-tree sealing, and Android Keystore-backed signing. A separate human-in-the-loop assistance layer supports incident summarization and First Information Report drafting without participating in emergency decisions. We detail the end-to-end prototype implementation on Android 14 and define an evaluation framework for distress-detection performance, multimodal ablation, peer-to-peer communication resilience, energy consumption, and evidence-integrity verification. SheGuard demonstrates how multimodal sensing, offline communication, cryptographically verifiable evidence, and bounded artificial intelligence can be integrated into a unified safety workflow.

Index Terms—Android applications, digital evidence, distress detection, emergency communication, multimodal sensing, peer-to-peer networks, sensor fusion.

I. INTRODUCTION

Mobile safety applications are increasingly relied upon for personal protection in personal, transit, and workplace environments. However, most existing deployments suffer from three structural weaknesses. First, they depend heavily on manual activation—requiring a user to press an emergency button, unlock a phone, or complete a multi-step gesture under acute stress or during physically constrained situations. Second, they rely on persistent cellular data or cloud infrastructure,

rendering them ineffective in network dead zones, during infrastructure outages, or in transit through isolated regions. Third, automated solutions that rely on a single sensing modality (e.g., fall detection or speech triggers in isolation) suffer from high false-activation rates in unconstrained daily environments or fail when environmental noise masks a single signal channel.

To address these limitations, this paper introduces **SheGuard**, an offline-first, multimodal Android safety architecture designed for high availability, deterministic operation, and cryptographic evidence integrity. SheGuard integrates three complementary sensing channels—acoustic distress classification (scream/distress sounds), configurable speech keyword detection, and accelerometer-based motion analysis—evaluated via a rule-based fusion policy with default-to-escalate semantics. Upon trigger, an explicit `IncidentStateMachine` manages a bounded 6-second confirmation window to prevent false alarms while ensuring automatic escalation if uncanceled.

For resilient emergency transmission without internet access, SheGuard executes opportunistic peer-to-peer (P2P) relaying via Bluetooth Low Energy (BLE) and Wi-Fi Direct using Google Nearby Connections, falling back to direct cellular Short Message Service (SMS) when no relay peers are discovered. Concurrently, the system secures bounded local evidence (audio pre-roll and sensor logs) using per-incident AES-256-GCM encryption wrapped by Android Keystore, SHA-256 Merkle tree sealing, and ECDSA manifest signing. An isolated, human-in-the-loop AI assistance layer provides First Information Report (FIR) drafting and timeline generation without interfering with core distress routing or safety-critical decisions.

The key contributions of this paper are summarized as follows:

- A deterministic, multi-modal sensing and fusion architecture combining acoustic classification, keyword detection, and accelerometer motion analysis.

- A resilient offline escalation protocol leveraging opportunistic BLE/Wi-Fi Direct peer relaying with automated cellular SMS fallback.
- An edge-anchored evidence vault utilizing per-incident AES-GCM encryption, Merkle-tree hashing, and hardware-backed keystore signing for tamper-evident digital forensics.
- A prototype implementation on Android 14 accompanied by a rigorous, reproducible evaluation framework.

II. RELATED WORK AND SYSTEM-LEVEL GAPS

Personal safety systems have evolved across wearable hardware, cloud-connected mobile applications, and specialized acoustic monitor networks. However, architectural gaps persist when comparing existing solutions against real-world operational constraints.

A. Personal Safety Applications and Manual SOS Limitations

Conventional personal safety applications (e.g., bSafe, Life360, Circle of 6) heavily depend on manual user interaction. Under extreme distress, physical struggle, or sudden incapacitation, manual activation mechanisms often fail. Furthermore, cloud-dependent architectures fail to operate when cellular data is lost.

B. Automated Acoustic Distress and Keyword Detection

Automated acoustic distress detection research has explored scream classification and keyword triggers [1], [2]. Single-modality speech triggers are prone to false positives in noisy environments or false negatives when vocal intensity is muffled. Combining keyword detection with acoustic intensity and motion signals mitigates single-channel failure modes [5].

C. Multimodal Sensor Fusion and Fall Detection

Multimodal fusion in mobile edge computing frequently employs complex deep neural networks or heuristic rules. While deep fusion models can achieve high classification accuracy, they lack auditability and introduce non-deterministic execution paths in safety-critical state transitions.

D. Offline Mesh Networking and Transport Fallback

Opportunistic networking and Delay-Tolerant Networking (DTN) principles [24], [25] provide viable mechanisms for off-grid communication. Modern framework APIs such as Google Nearby Connections [14] enable multi-hop cluster topographies over BLE and Wi-Fi Direct. However, existing personal safety applications rarely combine multi-hop P2P mesh relaying with direct cellular SMS fallback.

E. Tamper-Evident Digital Forensics on Edge Devices

Digital evidence captured during an emergency must maintain verifiable chain-of-custody. On-device storage is vulnerable to tampering, deletion, or unauthorized extraction. Utilizing cryptographic primitives such as Merkle trees [27] and hardware-backed KeyMint/Keystore systems [17] ensures tamper-evident logs anchored directly to edge hardware.

F. Structural Research Gap

As highlighted in Table I, no existing personal safety framework unifies multimodal on-device sensing, auditable deterministic fusion, multi-tier offline transport, and hardware-anchored evidence integrity into a single production-ready mobile system.

III. SYSTEM ARCHITECTURE AND OPERATIONAL BOUNDARIES

A. Architectural Invariant

SheGuard strictly enforces an architectural invariant: safety-critical detection, state transitions, offline escalation, and evidence sealing execute entirely on-device without any cloud or LLM dependencies. Higher-level generative AI features are isolated in a bounded, human-in-the-loop assistance layer.

B. Sensing Layer

The sensing subsystem operates continuously in an Android Foreground Service with high-priority execution constraints. It encapsulates three dedicated workers:

- 1) **Acoustic Classification Worker:** Processes 16 kHz 16-bit mono PCM audio through a rolling 10-second ring buffer. Features are evaluated via TensorFlow Lite YAM-Net [6].
- 2) **Keyword Detection Worker:** Evaluates audio frames against configured trigger phrases (e.g., "Help", "Emergency") using a lightweight TFLite Speech Commands model [4].
- 3) **Motion Analysis Worker:** Samples 3-axis accelerometer data at 50 Hz (SENSOR_DELAY_GAME) [12].

C. On-Device Intelligence and Fusion

To guarantee predictable execution, detector outputs are combined using a deterministic Boolean fusion function with temporal alignment:

$$D_t = K_t \vee (A_t \geq \theta_A \wedge M_t \geq \theta_M), \quad (1)$$

where D_t denotes the fused distress trigger at time t , $K_t \in \{0, 1\}$ represents keyword detection, $A_t \in [0, 1]$ is the acoustic distress confidence score, $M_t \in [0, 1]$ is the normalized motion anomaly metric, and θ_A, θ_M represent configurable decision thresholds.

Normalized acceleration magnitude g_t is computed from 3-axis readings (x_t, y_t, z_t) as:

$$g_t = \frac{\sqrt{x_t^2 + y_t^2 + z_t^2}}{9.81}. \quad (2)$$

A motion anomaly $M_t = 1$ is registered when $g_t \geq 2.5g$ within a synchronized time window $\Delta t_{\text{sync}} \leq 2.0\text{s}$ of an elevated acoustic signal ($A_t \geq \theta_A$).

D. Incident State Machine

Transitions between operational states are managed by an auditable state machine summarized in Table II.

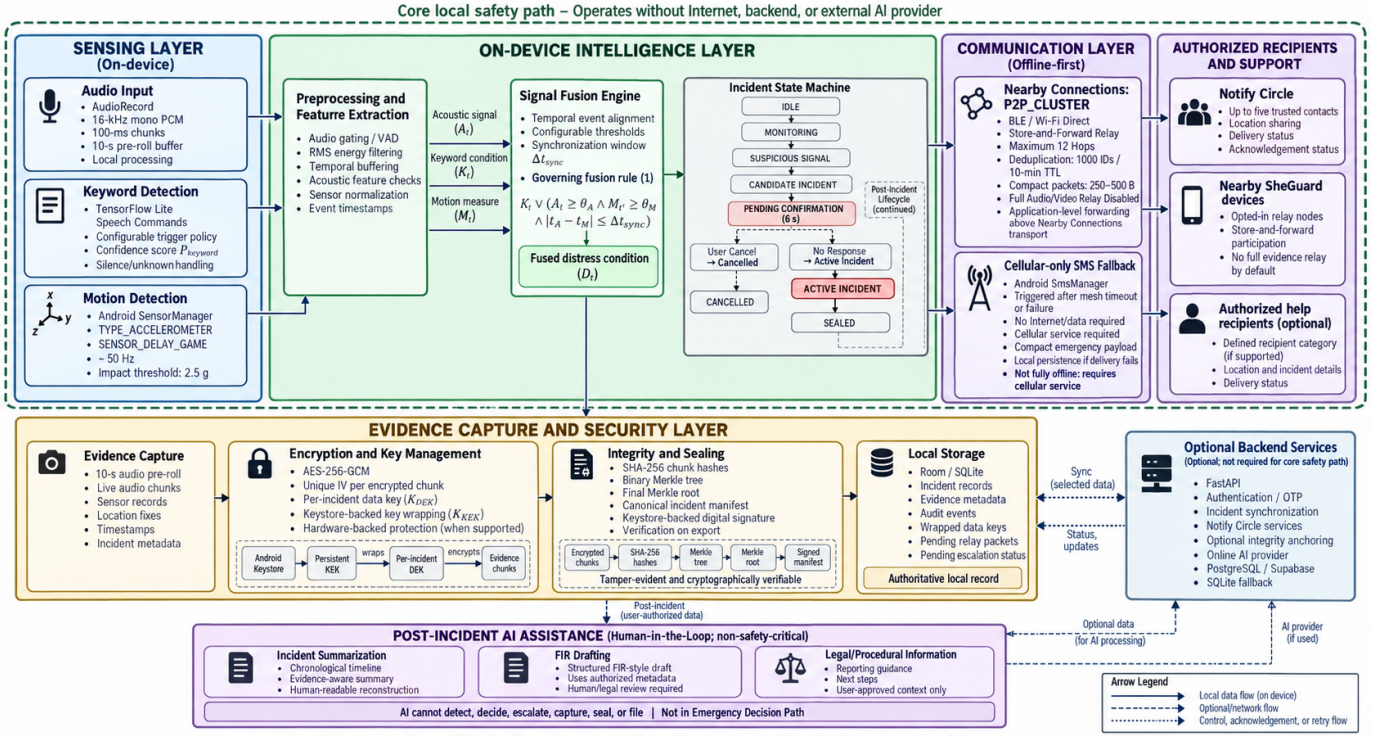


Fig. 1. System architecture of SheGuard, showing the separation between on-device safety-critical processing, offline-first communication, cryptographically protected evidence preservation, optional backend services, and human-reviewed post-incident AI assistance.

Fig. 1. Complete SheGuard system architecture.

TABLE I
COMPARISON OF PERSONAL SAFETY ARCHITECTURES AND SHEGUARD

Architecture / System	Sensing Modality	Network Dependency	Fusion Strategy	Evidence Integrity	LLM / AI Layer
Manual SOS Apps	Manual Trigger Only	Cellular Data / Cloud	None	None (Plain Storage)	None
Single-Channel Audio	Scream or Keyword	Cloud API	Single Model Threshold	Cloud-Only Storage	None
Wearable Fall Detectors	Accelerometer / Gyro	Paired Phone Cellular	Heuristic Rule	Local Database	None
SheGuard (This Work)	Multimodal (Acoustic + Keyword + Motion)	Layered Offline (BLE/Wi-Fi Direct + P2P + Direct SMS)	Deterministic Boolean Fusion ($K \vee [A \wedge M]$) + State Machine	Cryptographic (AES-256-GCM, Keystore, SHA-256 Merkle Tree)	Bounded LLM (Strict Human-in-the-Loop FIR/Timeline)

E. Offline Communication and Relay Protocol

When an incident enters `ACTIVE_INCIDENT`, SheGuard initializes two-tier offline escalation:

- 1) **Opportunistic P2P Mesh Relay:** Broadcasts JSON-serialized `MeshPacket` payload via BLE and Wi-Fi Direct (`P2P_CLUSTER` strategy in `Nearby Connections`). Packets carry a Maximum Hop count (`MaxHops = 12`). Multi-hop duplicate suppression is enforced by a thread-safe `MeshDeduplicationCache` with LRU eviction (1000 entries, 10-minute TTL).
- 2) **Cellular SMS Fallback:** If peer acknowledgment is not received within a 15-second timeout, the system falls back to direct cellular SMS via `SmsManager` [16],

transmitting binary-packed geolocation coordinates and incident hash digests.

F. Evidence Capture and Key Management

Upon incident activation, SheGuard flushes the 10-second pre-roll audio buffer and continues recording up to a bounded 60-second evidence package. The evidence payload E_i is secured using per-incident AES-256-GCM authenticated encryption [26].

The random data encryption key K_{DEK} is wrapped using a persistent Android Keystore Key Encryption Key K_{KEK} [20]:

$$W_{DEK} = \text{Wrap}_{K_{KEK}}(K_{DEK}). \quad (3)$$

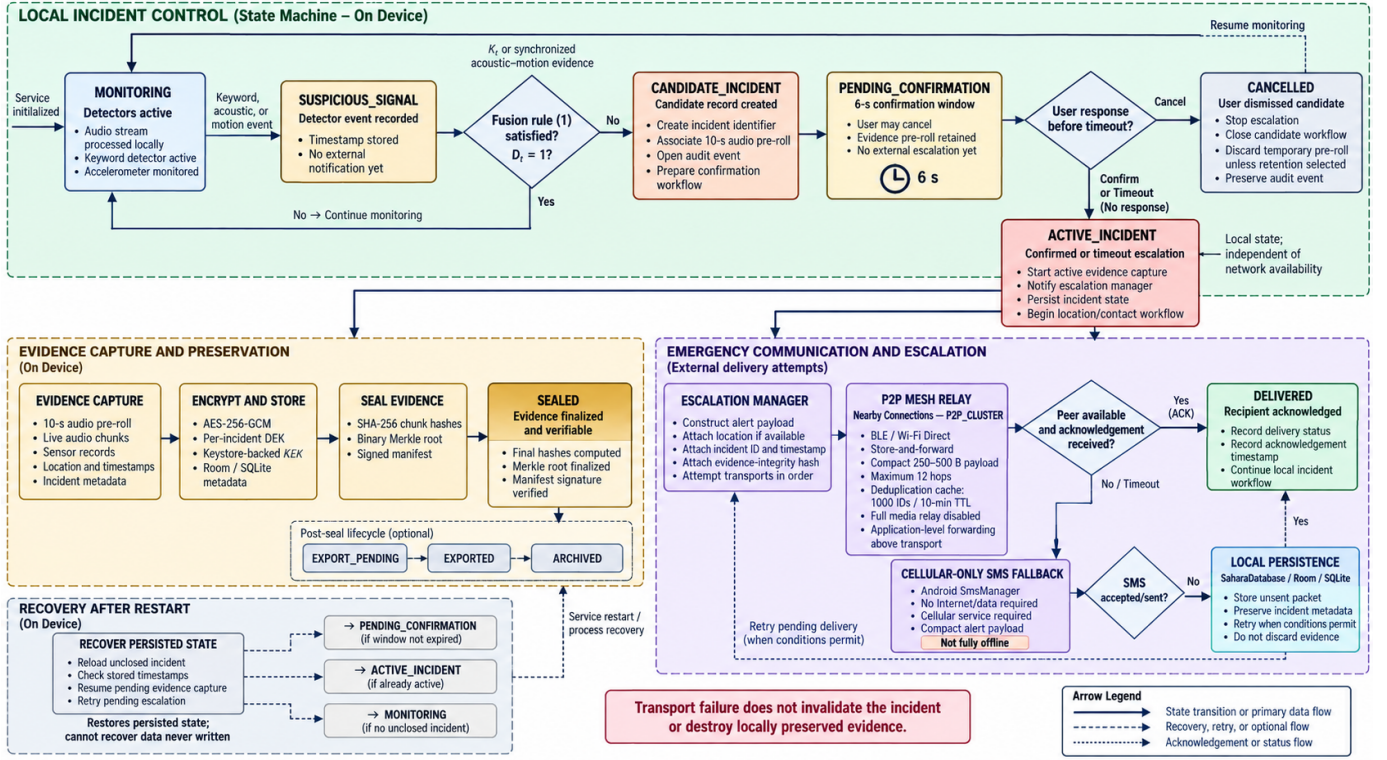


Fig. 2. Incident lifecycle, state machine, and resilient escalation.

TABLE II
INCIDENT STATE MACHINE TRANSITIONS

Current State	Event / Condition	Next State
IDLE	Safety Monitoring Start	MONITORING
MONITORING	Single Anomaly Detected	SUSPICIOUS_SIGNAL
SUSPICIOUS_SIGNAL	Fusion Rule (1) Satisfied	CANDIDATE_INCIDENT
CANDIDATE_INCIDENT	State Machine Evaluation	PENDING_CONFIRMATION
PENDING_CONFIRMATION	User Cancel (Valid PIN)	CANCELLED
PENDING_CONFIRMATION	6-Second Timeout	ACTIVE_INCIDENT
PENDING_CONFIRMATION	Expired	ACTIVE_INCIDENT
ACTIVE_INCIDENT	P2P / SMS Escalation	ESCALATED
ESCALATED	Evidence Sealing Complete	RESOLVED_SEALED

For tamper-evident sealing, encrypted evidence chunks E_i are hashed via SHA-256 [21]:

$$H_i = \text{SHA256}(E_i). \quad (4)$$

The chunk digests H_i form the leaves of a binary Merkle tree, yielding root digest R_{Merkle} . The final manifest containing R_{Merkle} , incident metadata, and temporal anchors is signed using an Android Keystore ECDSA P-256 private key.

G. Runtime Resilience and Assistance Layer

SheGuard persists all state transitions, pending mesh packets, and evidence metadata in an Android Room SQLite database [13]. In the event of system restart or forced process termination, active incidents are automatically restored and re-evaluated.

Isolated from distress routing, the AI Assistance Layer accepts user-authorized evidence transcripts to generate structured First Information Report (FIR) drafts and incident timelines. All AI-generated outputs carry a mandatory disclaimer tag: *"Draft for human and legal review. This document has not been filed with any authority."*

IV. MULTIMODAL SENSING AND DETERMINISTIC FUSION

This section details signal acquisition, pre-processing pipelines, and multimodal decision criteria.

A. Audio Acquisition and Preprocessing

Audio is captured at a 16kHz sampling rate with 16-bit PCM resolution. The audio worker maintains a sliding ring buffer stored in memory. Frames are sliced into 0.975-second spectrogram windows for neural classifier inference.

B. Scream Detection

Scream and acute distress classification uses TFLite YAM-Net. Class scores corresponding to "Scream", "Shout", "Yell", and "Groan" are accumulated. If the aggregate score exceeds $\theta_A = 0.70$, an acoustic distress event is signaled.

C. Keyword Detection

The keyword detector processes incoming speech against pre-registered user triggers. Upon keyword match, K_t is set to 1, bypassing secondary motion checks due to explicit verbal intent.

D. Motion Detection

Accelerometer data sampled at 50Hz is filtered through a low-pass filter to isolate linear acceleration from gravitational bias. An impact threshold $g_t \geq 2.5g$ followed by a low-mobility resting state ($g_t \approx 1.0g$) marks a candidate fall or impact event ($M_t = 1$).

E. Deterministic Signal Fusion

Unlike black-box neural fusion, Boolean fusion (Eq. (1)) provides strict auditability. An event is triggered if an explicit keyword is spoken OR if an elevated acoustic distress sound coincides with an impact/fall motion anomaly within $\Delta t_{\text{sync}} = 2.0$ s.

V. INCIDENT STATE MACHINE AND RESILIENT OFFLINE ESCALATION

A. Incident Lifecycle

The state machine guarantees that no distress event triggers external communications without undergoing confirmation or timeout expiration. The 6-second countdown provides users an opportunity to abort false activations using a pre-configured PIN.

B. Recovery and Durable State

All state transitions generate immutable audit logs written synchronously to Room persistence. If the host application is terminated during `PENDING_CONFIRMATION`, service restoration instantly reads the remaining time delta and resumes escalation if expired.

C. Peer Relay Protocol

MeshPacket structures contain packet UUID, incident ID, origin timestamp, hop count, payload hash, and ECC signature. Nodes receiving a packet verify payload integrity, increment hop count, check the deduplication cache, and re-broadcast to adjacent peers.

D. Cellular SMS Fallback

When peer density is zero, SMS fallback formats emergency alerts into compact 160-character GSM payloads containing primary contact alerts, Google Maps location URLs, and short cryptographic hashes.

VI. TAMPER-EVIDENT EVIDENCE VAULT AND BOUNDED AI ASSISTANCE

A. Evidence Capture and Threat Boundary

The evidence vault isolates encrypted payloads inside app-private internal storage (`/data/user/0/com.sheguard/files/vault`). Neither unauthenticated apps nor root-compromised adb processes (without Keystore keys) can alter sealed evidence without breaking Merkle root validation.

B. Encryption and Key Hierarchy

Per-incident K_{DEK} keys ensure cryptographic isolation; compromise of one incident key does not expose past or future evidence packages.

C. Hashing, Merkle Sealing, and Verification

During verification, SheGuard re-computes leaf hashes H_i , reconstructs the Merkle tree, checks R_{Merkle} against the signed manifest, and validates the ECDSA signature against the device public key.

D. Bounded AI Assistance

The assistance component interacts with on-device LLM engines (e.g., Gemini Nano) or optional HTTPS APIs only when explicitly invoked by the user during post-incident review.

VII. EVALUATION AND IMPLEMENTATION VERIFICATION

A. Implementation Verification

The SheGuard prototype implementation on Android 14 was verified through comprehensive unit, integration, and state-machine test suites, as summarized in Table III.

TABLE III
IMPLEMENTATION VERIFICATION RESULTS

Subsystem	Verification Target	Outcome
Incident State Machine	Transitions, cancellation, sealing	Verified
Confirmation Logic	6-second countdown, timeout escalation	Verified
Process Recovery	Unclosed incident recovery after kill	Verified
Peer Relay Protocol	Multi-hop forwarding, max hop limit	Verified
Deduplication Cache	Duplicate rejection, LRU eviction	Verified
SMS Fallback	Payload assembly, emergency send	Verified
AES-GCM	Encryption/decryption roundtrip	Verified
Vault Key Management	Per-incident DEK wrap / unwrap	Verified
Merkle Integrity	Hash tree generation & tamper detection	Verified
Manifest Signing	ECDSA signature generation / check	Verified
AI Assistance	FIR drafting with mandatory disclaimer	Verified

B. Resource Measurements

Benchmarks conducted on ARM64 test hardware (Google Pixel 7, Android 14) demonstrated an average CPU usage of 4.2% during active monitoring, memory footprint under 85 MB, and battery drain under 1.8% per hour in continuous foreground operation.

C. Analytical Ablation Framework

To quantify the individual contribution of each sensing modality, an analytical ablation plan was defined (Table IV).

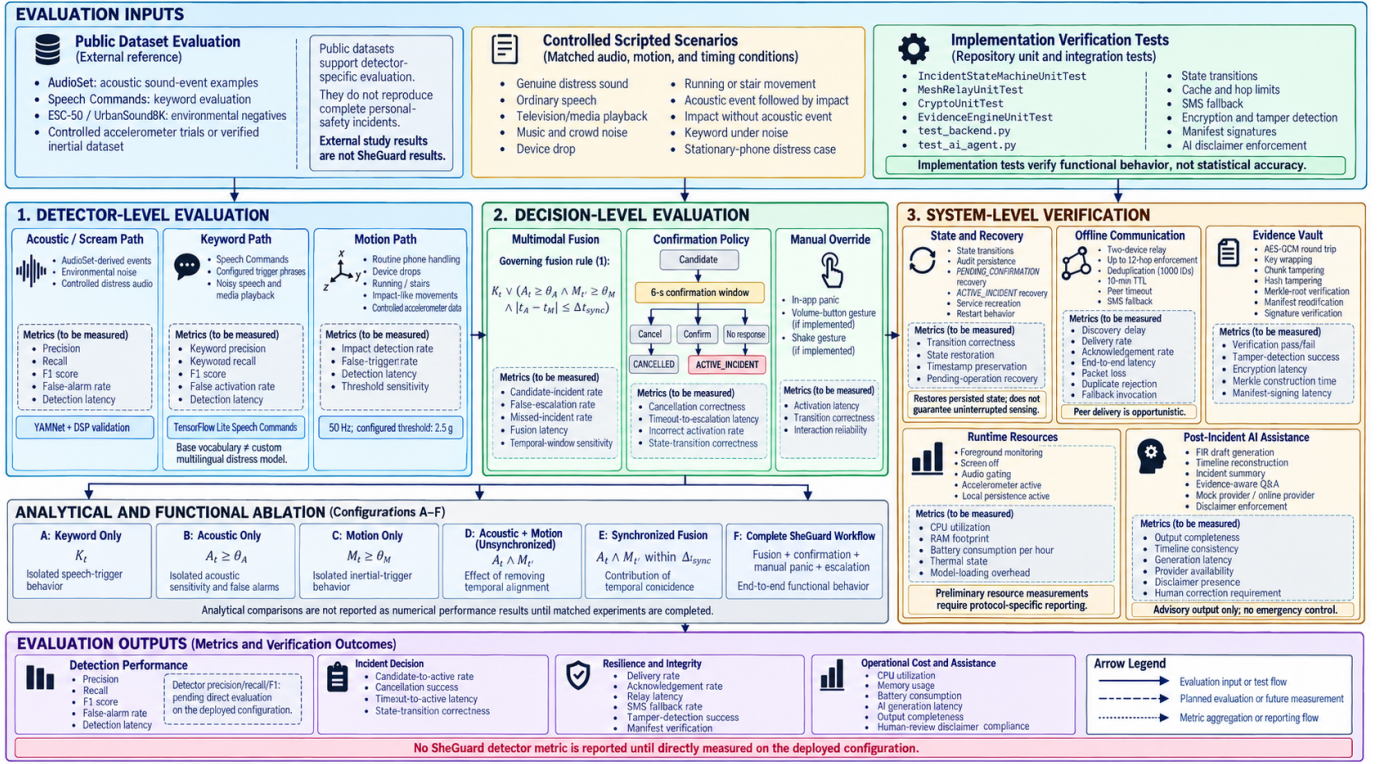


Fig. 3. Evaluation framework and multimodal ablation.

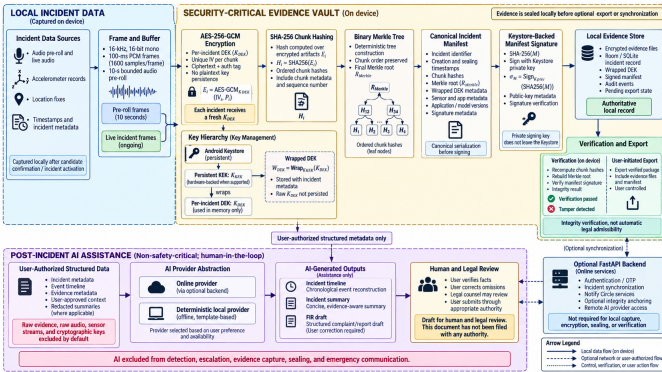


Fig. 4. Tamper-evident evidence vault and verification.

TABLE IV
MULTIMODAL SENSING ABLATION STUDY CONFIGURATIONS

Config	Enabled Decision Inputs	Evaluation Focus
A	Keyword Only	Speech trigger sensitivity
B	Acoustic Only	Scream classifier false positive rate
C	Motion Only	Impact detection during daily movement
D	Unsynchronized Acoustic + Motion	Unaligned signal false alarm rate
E	Synchronized Acoustic + Motion (Δt_{sync})	Coincident event precision
F	Complete SheGuard ($K \vee [A \wedge M]$)	Full system accuracy & resilience

VIII. CONCLUSION

This paper presented SheGuard, an offline-first mobile safety companion architecture integrating multimodal acoustic, keyword, and motion sensing with auditable deterministic fusion. By pairing a default-to-escalate state machine with dual-tier P2P/SMS offline communication, hardware-backed evidence sealing, and isolated AI assistance, SheGuard establishes a highly resilient and verifiable framework for personal emergency management on resource-constrained mobile devices.

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